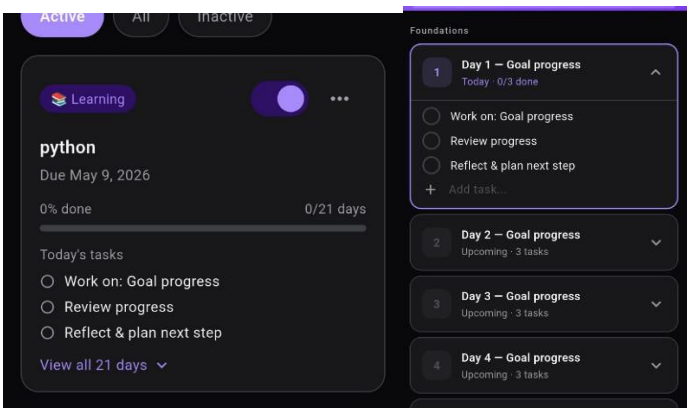
LAB 8

Navigator, Hero Animation, BottomNavBar

1. NAVIGATOR:

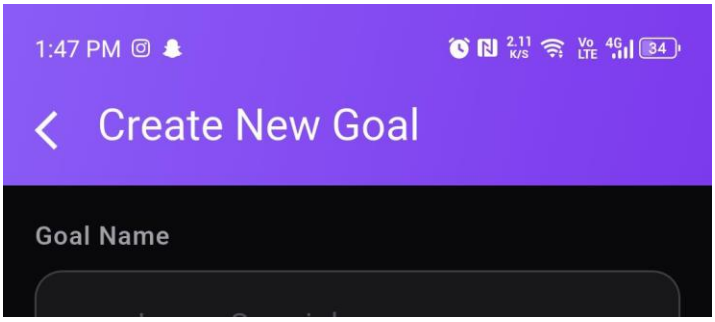
Goal screen:

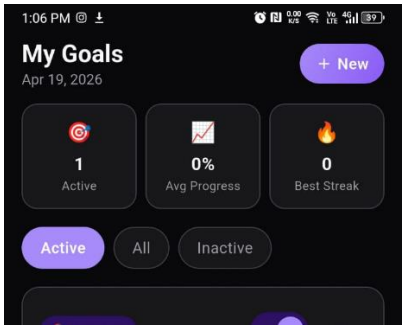
```
onTap: () => Navigator.push(
  context,
  MaterialPageRoute(builder: (_) => GoalDetailScreen(goal: goal)),
),
```



Goal Details Screen:

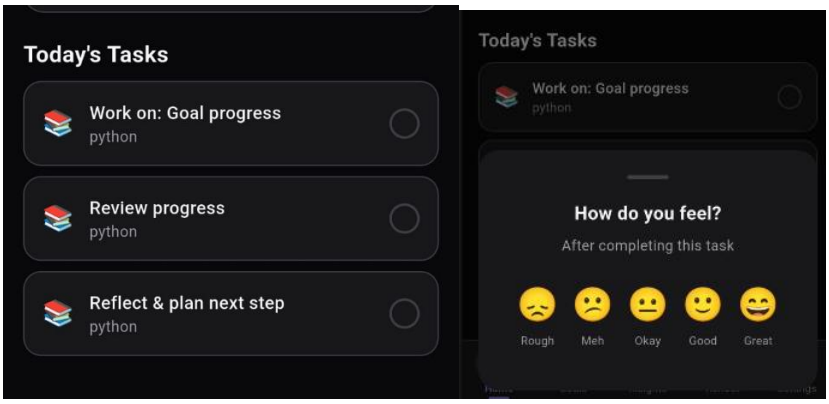
```
leading: IconButton(
  icon: const Icon(Icons.arrow_back_ios, color: Colors.white),
  onPressed: () => Navigator.pop(context),
),
```

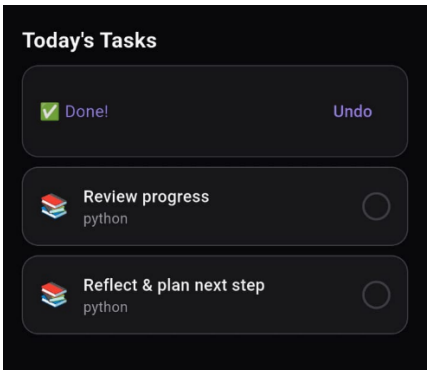




Home screen

```
536            return GestureDetector(
537              onTap: () => Navigator.pop(context, i),
538              child: Column(
539                children: [
540                  Text(
541                    MoodEntry.moodEmojis[i],
542                    style: const TextStyle(fontSize: 32),
543                  ),
```





Navigator Push Pop:

- **Push:** new screen slides in from right.
- **Pop:** current screen slides out to right, revealing previous screen.
- Both Considered Standard Android/Material navigation pattern.
- Tap goal card pushes the GoalDetailScreen on top slides in from the right.
- White back arrow in the goal detail header tapping it slides screen back off to the right.
- Mood picker tapping an emoji pops the modal AND returns the selected mood index back to the caller.

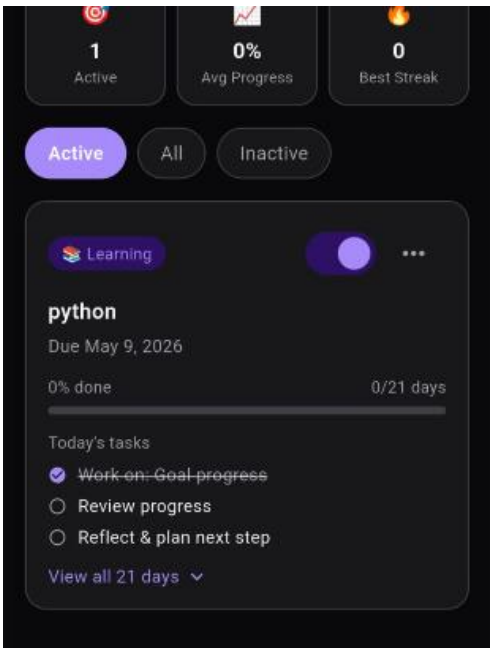
2. HERO ANIMATION

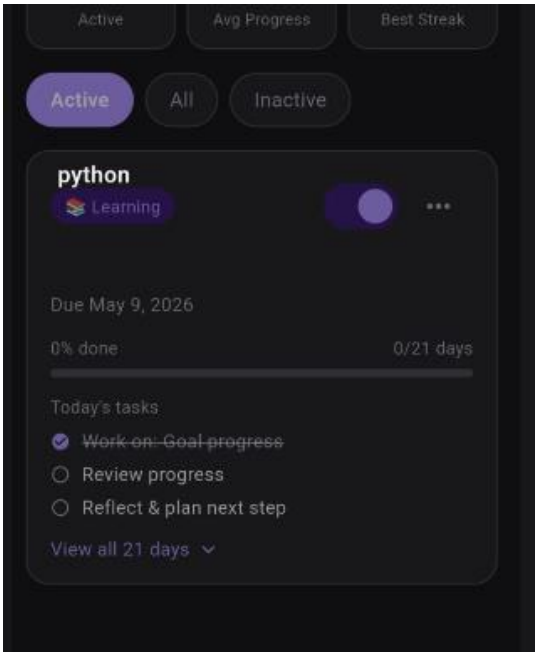
Goal_screen

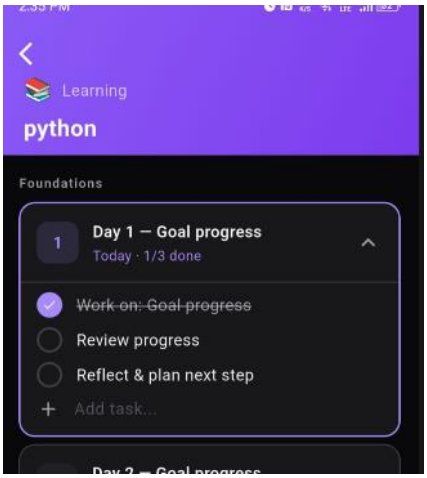
```
child: Hero(
  tag: 'goal-name-${goal.id}',
  child: Material(
    color: Colors.transparent,
    child: Text(
      goal.name,
      style: TextStyle(
        color: theme.text,
        fontSize: 16,
        fontWeight: FontWeight.w600,
      ),
    ),
  ),
),
```

Goal_details_screen

```
Hero(
  tag: 'goal-name-${goal.id}',
  child: Material(
    color: Colors.transparent,
    child: Text(
      goal.name,
      style: const TextStyle(
        color: Colors.white,
        fontSize: 20,
        fontWeight: FontWeight.bold,
      ),
    ),
  ),
),
```







Hero Animation:

- Tap any goal in Goals tab the goal
- **Title text flies** from the card (dark, 16px) to the detail screen header (white, 20px)
- Smoothly morphing size, color and position mid-flight.
- Flutter matches the two Hero widgets by identical tag and animates between them automatically.

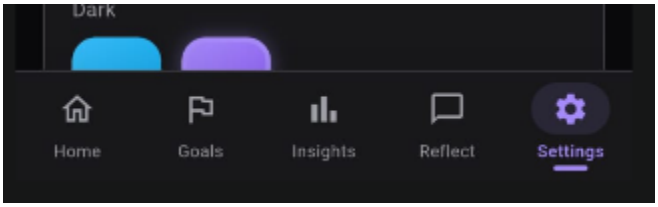
3. BOTTOM NAVIGATION BAR

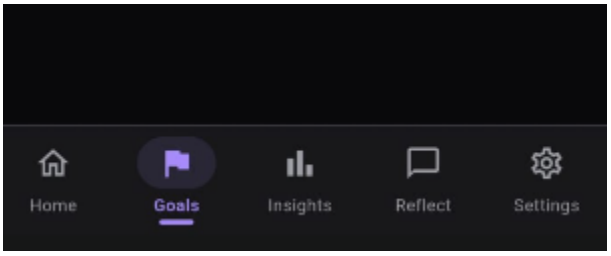
Main screen

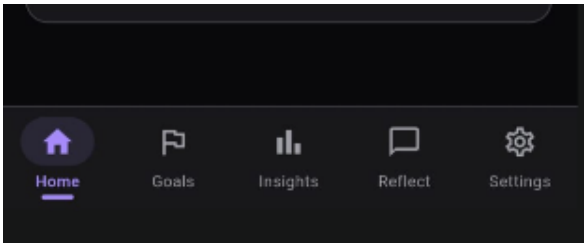
```
final _tabs = const [
  _TabItem(icon: Icons.home_outlined, activeIcon: Icons.home, label: 'Home'),
  _TabItem(icon: Icons.flag_outlined, activeIcon: Icons.flag, label: 'Goals'),
  _TabItem(icon: Icons.bar_chart_outlined, activeIcon: Icons.bar_chart, label:
  'Insights'),
  _TabItem(icon: Icons.chat_bubble_outline, activeIcon: Icons.chat_bubble, label:
  'Reflect'),
  _TabItem(icon: Icons.settings_outlined, activeIcon: Icons.settings, label:
  'Settings'),
];
```

```
      bottomNavigationBar: Container(
        decoration: BoxDecoration(
          color: theme.card,
          border: Border(top: BorderSide(color: theme.border)),
          boxShadow: [
            BoxShadow(
              color: Colors.black.withValues(alpha: 0.08),
              blurRadius: 20,
              offset: const Offset(0, -4),
            ),
          ],
        ),
        child: SafeArea(
          child: SizedBox(
            height: 64,
            child: Row(
              children: List.generate(_tabs.length, (i) {
                final tab = _tabs[i];
                final active = i == _currentIndex;
                return Expanded(
                  child: GestureDetector(
                    behavior: HitTestBehavior.opaque,
                    onTap: () => setState(() => _currentIndex = i),
                    child: Column(
                      mainAxisAlignment: MainAxisAlignment.center,
                      children: [
                        AnimatedContainer(
                          duration: const Duration(milliseconds: 200),
                          padding: const EdgeInsets.symmetric(
                              horizontal: 12, vertical: 4),
                          decoration: BoxDecoration(
                            color: active
                                ? theme.accent.withValues(alpha: 0.12)
                                : Colors.transparent,
                            borderRadius: BorderRadius.circular(20),
                          ),
```

```
                    child: Icon(
                      active ? tab.activeIcon : tab.icon,
                      color: active ? theme.accent : theme.textMuted,
                      size: 22,
                    ),
                  ),
                  const SizedBox(height: 2),
                  Text(
                    tab.label,
                    style: TextStyle(
                      fontSize: 10,
                      color: active ? theme.accent : theme.textMuted,
                      fontWeight: active
                          ? FontWeight.w600
                          : FontWeight.normal,
                    ),
                  ),
                  const SizedBox(height: 2),
                  AnimatedContainer(
                    duration: const Duration(milliseconds: 200),
                    height: 3,
                    width: active ? 20 : 0,
                    decoration: BoxDecoration(
                      color: theme.accent,
                      borderRadius: BorderRadius.circular(2),
                    ),
                  ),
                ],
              ),
            ),
          );
        }),
      ),
```







Bottom Navigation Bar:

- Bottom bar with 5 tabs.
- Active tab = filled icon + accent color + light background pill + small underline dot (all animated 200ms).
- Inactive tabs = outlined icon + muted gray.
- Tapping any tab instantly switches the screen content using IndexedStack (keeps state of all screens alive).